



Gaetano Settembre

Employeneur - TMC Italia S.p.A.

AI/ML Engineer c/o Leonardo S.p.A. - Space Division

PhD in Computer Science and Mathematics | MSc in Data Science

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📍 Gravina in Puglia (BA) - 70024, IT

📍 Campi Bisenzio (FI) - 50013, IT

WORK AND ACADEMIC EXPERIENCE

• TMC Italia S.p.A.

Dec 2025 - Present

Engineering Consultant: AI/ML Engineer c/o Leonardo SpA - Space Division

Campi Bisenzio/Milan, Italy

- Development and validation of **AI/ML-based on-board processing solutions for innovative satellite payloads**, including end-to-end processing pipelines for multispectral and hyperspectral Earth Observation (EO) data, covering corrections, cloud detection, semantic segmentation, and object detection.
- **Performance, power consumption, and timing assessment of AI models** and conventional algorithms on representative flight hardware, optimizing computational efficiency and resource utilization on Unibap iX10 platforms (AMD GPU, Intel Myriad X or Hailo-8 accelerators).
- Training, validation, and maintenance of **deep learning models in HPC multi-GPU environments for Very High Resolution (VHR) instrument simulators**, supporting advanced processing tasks such as image refocusing and denoising.
- Contribution to the **definition of space segment and payload system requirements**, focusing on EO instruments and on-board processing units to ensure alignment between mission objectives, user needs, and technical constraints.

• Planetek Italia S.r.l.

Nov 2022 - Sep 2023

Visiting Researcher

Bari, Italy

- Collaborated with research and engineering teams to integrate data-driven and ML-based approaches for Earth Observation applications (wildfire segmentation, plastic litter detection, etc).
- Designed and developed methodologies for the analysis of PRISMA (PRecursore IperSpettrale della Missione Applicativa) hyperspectral imagery [C.4, C.5].
- Contributed to the definition, prototyping, and implementation of a land cover change detection framework in wildfire-affected areas using PRISMA hyperspectral data [J.2].

• Institute of Crystallography - CNR

Oct 2021 - Mar 2022

Academic Internship

Bari, Italy

- Designed and implemented ETL processes for crystallographic data ingestion from open-source databases.
- Trained, evaluated, and optimized machine learning models for crystal system classification (multi-class task).
- Contributed to the deployment of trained models and the development of the web platform *CrystalMELA* (Crystallography MachinE LeArning) [J.4].

• SER&Practices S.r.l.

Nov 2018 - Apr 2019

Academic Internship

Bari, Italy

- Designed and developed "serious games" based on the Red Team vs. Blue Team approach.
- Conducted penetration testing and vulnerability assessments to identify and mitigate security flaws.
- Performed data collection and analysis (e.g., logs, alerts), producing security reports using IBM QRadar SIEM.

EDUCATION

- **PhD in Computer Science and Mathematics**

Oct 2022 - Oct 2025

University of Bari Aldo Moro, Department of Mathematics

Bari, Italy

- PhD fellowship awarded under D.M. 352/22 in collaboration with Planetek Italia S.r.l.
- Research topics: low-rank approximation, NMF variants, spatial regularization, low-dimensional learning models (DeepNMF, Autoencoders), and Earth Observation applications (e.g., Land cover classification, wildfire segmentation, Plastic litter detection, etc).
- Thesis title: *"Low-dimensional Learning Models for Hyperspectral Data Analysis in Environmental Monitoring"*.
- Final grade: **Excellent**

- **Master's Degree in Data Science (LM-91)**

Oct 2019 - Mar 2022

University of Bari Aldo Moro, Department of Computer Science

Bari, Italy

- Final grade: **Full mark with honors - 110/110 & Lode** (highest possible grade in the Italian system).
- Thesis in Decision Models and Optimization *"Analisi di dati cristallografici e costruzione di modelli predittivi"*.

- **Bachelor's Degree in Computer Science (L-31)**

Sep 2015 - Apr 2019

University of Bari Aldo Moro, Department of Computer Science

Bari, Italy

- Final grade: **107/110**
- Thesis in Software Engineering and Cyber Security: *"Red Team vs Blue Team: Data Exfiltration in Government"*. Supervisor: Prof. D. Caivano.

SKILLS

- **Core Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, spaCy, NLTK, Knime, Weka, Orange, giotto-tda, XAI tools (e.g., SHAP, Grad-CAM)
- **Programming Languages:** Python, MATLAB, R
- **Database Systems:** MySQL, PostgreSQL + PostGIS, GeoDB
- **DevOps & Version Control:** Git, GitHub, GitLab CI/CD, Docker
- **Geospatial tools:** QGIS, xarray, geopandas, rasterio, Microsoft Planetary Computer, SNAP ESA, ENVI
- **Mathematical & Statistical Tools:** MATLAB Toolboxes, SciPy, R (tidyverse, caret)
- **Web Technologies:** HTML, CSS, JavaScript
- **Other Tools & Technologies:** IBM QRadar, IBM Eng. Requirements Management DOORS
- **Research Skills:** Scientific writing, Literature and peer review, Experimental design, Data preprocessing, Model evaluation, Reproducible research

SELECTED COURSES AND CERTIFICATIONS

All badges of verified achievements are available at the following link: [Credly profile](#) or LinkedIn.

- **Summer School**, 4th CINI International School on High-Performance Computing (HPC) Jun 2026
- **Anthropic**, AI Fluency Framework & Foundations Mar 2026
- **Doctoral Winter School**, Generative AI and Actions for a changing world (GAIA2025) Nov 2025
- **Summer School**, 6th Advanced Course on Data Science & Machine Learning (ACDL 2023) Jun 2023
- **Spring School**, DeepLearn - 9th International School on Deep Learning Apr 2023
- **Politecnico of Bari**, Satellites Systems for Remote Sensing and Geolocation (6 CFU) Oct 2022 - Jan 2023
- **Planetek Italia**, Processing of Remote Sensing Data Oct 2022

ACADEMIC HIGHLIGHTS

All publications, reviews, and further information can be found on [Google Scholar](#), [ORCID](#), and the [personal website](#), as well as on LinkedIn profile.

- **Author of 10+ peer-reviewed scientific publications** (4 journal articles, 7 conference papers) in Machine Learning and Artificial Intelligence, Computer Vision for Remote Sensing, Geoscience, Medical Imaging, and Data Analysis.
- **Speaker at 8+ national and international scientific events**, including *ECML-PKDD 2023*, *IEEE IGARSS 2024*, *KES 2025*, *NASCA 2023*, and *PhD Days*.
- **Co-organizer of 10 scientific events** (international workshops, special sessions, technical tracks) within IEEE, ACM, KES, SIMAI, and national research communities and international conferences.
- **Reviewer for 12+ international journals** including *IEEE Trans. on Geoscience and Remote Sensing*, *Scientific Reports*, *Scientific Data*, *Biomedical Signal Processing and Control*, *Journal of Systems and Software*, and **5+ international conferences** including *IEEE IJCNN 2025*, *ICIAP 2025*, *KES 2024-2025*, *ACM SAC 2026*, *ISCC 2026*, and *IEEE ICTS4eHealth 2026*.
- **Contributor to national and international funded research projects** involving Earth Observation, Health Informatics, and Data Science in applied sciences.

ADDITIONAL INFORMATION

- **Languages:** Italian (Mother tongue), English (B1 - Intermediate level)
- **Basketball Referee (FIP):** Referee of the Italian Basketball Federation (Federazione Italiana Pallacanestro), License No. A056193.
- **Driving licence:** AM, B

May 27, 2026

Gaetano Settembre

DETAILED ACADEMIC INFORMATION

PARTICIPATION IN NATIONAL AND INTERNATIONAL RESEARCH PROJECTS

- **MODA: Model-based and Data-Driven Methods for Multiscale Biological Systems** *Mar 2025 - Mar 2026*
Coordinator: Assist. Prof. Marta Menci (University Campus Biomedico Rome, Italy)
 - Funded by Gruppo Nazionale Calcolo Scientifico - Istituto Nazionale di Alta Matematica (GNCS-INdAM). CUP: E53C24001950001.
- **MBI: Marine Biomass Innovation** *Oct 2022 - Feb 2026*
Principal Investigator: Prof. Raymond Thomas (Western University Ontario, Canada)
 - Funded by New Frontiers in Research Fund - Transformation Stream, Canada. NFRFT-2020-00452.
- **Low-rank models and Optimization algorithms for Data Analysis** *Mar 2024 - Mar 2025*
Coordinator: Prof. Margherita Porcelli (University of Florence, Italy)
 - Funded by Gruppo Nazionale Calcolo Scientifico - Istituto Nazionale di Alta Matematica (GNCS-INdAM). CUP: E53C23001670001.

CONFERENCE AND WORKSHOP ORGANIZATION

- **INSIGHT: INtelligent Systems for Imaging-based diaGnosis in HealThcare** *Jun TBD, 2026. Algarve, PT*
International Special Session (1st Edition) 
 - At 6th IEEE International Conference on ICT Solutions for eHealth (ICTS4eHealth 2026)
- **NUMERA: advanced NUMerical MEthods for Real-world Applications** *Jun 15, 2026. Bari, IT*
National Workshop (1st Edition) 
- **KES: Knowledge-Based and Intelligent Information & Engineering Systems** *Sep 09-11, 2025. Dublin, IRL*
International Main Conference (30th Edition) - KES 2026 
- **ASH: Applications and Systems for Healthcare** *Mar 23-27, 2026. Thessaloniki, GR*
International Technical Track 
 - At 40th ACM/SIGAPP Symposium On Applied Computing (SAC 2026)
- **Innovative Medical image Processing with AI-driven preCision Technologies** *Sep 15, 2025. Rome, IT*
International Workshop (1st Edition) 
 - At 23rd International Conference on Image Analysis and Processing (ICIAP 2025)
- **SOAP4Earth: Smart Observation And Preservation for Earth** *Sep 10-12, 2025. Osaka, JP*
International Invited Session (1st Edition) 
 - At 29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025)
- **Advanced Numerical Methods and Machine Learning Techniques in Applied Science** *Sep 1, 2025. Trieste, IT*
International Minisymposium (1st Edition) 
 - At Biennial congress of the Italian Society of Applied and Industrial Mathematics (SIMAI 2025)
- **Mathematics for Artificial Intelligence and Machine Learning** *Jan 29-31, 2025. Bari, IT*
National UMI Group Workshop (3rd Edition) 
- **LDTH: Leveraging Digital Twins in Healthcare** *Sep 11-13, 2024. Seville, ES*
International Invited Session (1st and 2nd Edition) 
 - At 28 and 29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024, 2025)
- **PRIMO: Post-graduate Researches in Inverse Problems, ML, and Optimization** *Sep 20-22, 2023. Bari, IT*
National Workshop (3rd Edition) 

- [J.1] **G. Settembre**, F. Esposito, N. Del Buono (2025). Advancing blind hyperspectral unmixing in remote sensing: comparing deep-inspired subspace learning methods. *Adv. Modeling and Simulation in Eng. Sciences*, vol. 12, pp. 30.
- [J.2] **G. Settembre**, N. Taggio, N. Del Buono, F. Esposito, P. Di Lauro, A. Aiello (2024). A Land Cover Change Framework Analyzing Wildfire-affected Areas in Bitemporal PRISMA Hyperspectral Images. *Mathematics and Computers in Simulation*, vol. 229, pp. 855-866.
- [J.3] S.G. De Benedictis, G. Gargano, **G. Settembre** (2024). Enhanced MRI brain tumor detection and classification via topological data analysis and low-rank tensor decomposition. *J. of Comp. Mathematics and Data Science*, vol. 13, pp. 100103.
- [J.4] N. Corriero, R. Rizzi, **G. Settembre**, N. Del Buono, D. Diacono (2023). CrystalMELA: a new crystallographic machine learning platform for crystal system determination. *J. of Appl. Crystallography*, vol. 56, pp. 409-419.
- [C.1] C. Russo, **G. Settembre**, G. Gargano, M.S. de Biase, R. De Fazio (2026). AI in Pediatric Urology: Deep Learning-based Approach supporting Posterior Urethral Valves Diagnosis on VCUG Imaging. In *International Conference on Image Analysis and Processing (ICIAP 2025)*. Lecture Notes in Computer Science, vol. 16170, pp. 138-149.
- [C.2] **G. Settembre**, G. Gargano, N. Del Buono (2025). Superpixel-based plastic litter detection in UAV hyperspectral imaging using spectral-textural features. In *International Conference on Knowledge Based and Intelligent information and Engineering Systems (KES 2025)*. Procedia Computer Science, vol. 270, pp. 4997-5006.
- [C.3] G. Gargano, **G. Settembre**, N. Del Buono (2025). AI-Driven Insights into Microbial Biomarkers for Colorectal Cancer Progression. In *International Conference on Knowledge Based and Intelligent information and Engineering Systems (KES 2025)*. Procedia Computer Science, vol. 270, pp. 4936-4945.
- [C.4] **G. Settembre**, N. Del Buono, F. Esposito, N. Taggio (2024). Deep NMF and Autoencoder: A Comparative Analysis for Hyperspectral Unmixing Using PRISMA Real Images. In *IGARSS 2024 - IEEE International Geoscience and Remote Sensing Symposium*. pp. 3708-3712.
- [C.5] **G. Settembre**, N. Taggio, N. Del Buono, A. Aiello, F. Esposito (2023). Low-rank hierarchical clustering of PRISMA hyperspectral images to identify burned areas. In *European Conference on Machine Learning and Knowledge Discovery in Databases. (ECML-PKDD 2023)*. Communications in Computer and Information Science, vol. 2135, pp. 412-423.
- [C.6] **G. Settembre**, N. Corriero, N. Del Buono, F. Esposito, R. Rizzi (2023). Machine Learning approaches for Predicting Crystal Systems: A brief Review and a case study. In *Int. conference on Machine Learning, Optimization, and Data Science (LOD 2023)*. Lecture Notes in Computer Science, vol 13810. pp. 93-107.

PARTICIPATION AS SPEAKER AT NATIONAL AND INTERNATIONAL SCIENTIFIC MEETINGS

- **Oral Presentation** Sep 10-12, 2025, Osaka, JP
Int. Conference on Knowledge Based and Intelligent information and Engineering Systems (KES)
 - Talk title: " Superpixel-based plastic litter detection in UAV hyperspectral imaging using spectral-textural features"
- **Oral Presentation** Jul 15, 2025, Bari, IT
PhD Days 2023-2025
 - Talk title: "Updates on low-rank models for the analysis of Earth Observation data"
- **Poster Presentation** Jan 29-31, 2025, Bari, IT
Math4AIML - Workshop on Mathematics for Artificial Intelligence and Machine Learning
 - Poster title: "Spatial Informed Hierarchical Clustering for Hyperspectral Imagery via Total Variation"
- **Poster Presentation** Jul 11, 2024, Athens, GR
IGARSS 2024 - IEEE International Geoscience and Remote Sensing Symposium
 - Poster title: "Deep NMF and Autoencoder: A Comparative Analysis for Hyperspectral Unmixing Using PRISMA Real Images"
- **Oral Presentation** Sep 18, 2023, Turin, IT
Eur. Conf. on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)
 - Talk title: "Low-rank hierarchical clustering of PRISMA hyperspectral images to identify burned areas" in MACLEAN: Workshop on Machine Learning for Earth Observation
- **Invited Speaker** Dec 21, 2023, Bari, IT
XMaths Workshop, Ed. 2023
 - Talk title: "Unlocking insights from the sky: low-rank methods in remote sensing"
- **Oral Presentation** Jul 06, 2023, Athens, GR
Int. Conference on Numerical Analysis and Scientific Computation with Applications (NASCA)

- Talk title: "A low-rank based approximation framework for real world data integration"
- **Oral Presentation** *Sep 18, 2022, Castiglione della Pescaia, IT*
Int. Conference on Machine Learning, Optimization and Data science (LOD)
 - Talk title: "Advanced Machine Learning approaches for Predicting Crystal Systems"

PROFESSIONAL MEMBERSHIPS

- **KBIES**: Knowledge-Based and Intelligent Engineering Systems Society member *Mar 2025 - Present*
- **INdAM-GNCS**: Gruppo Nazionale Calcolo Scientifico member *Sep 2023 - Present*
- **PRIMO**: Post graduate Researchers in Inverse problems, ML and Optimization member *Dec 2022 - Present*
- **SIMAI**: Italian Society of Applied and Industrial Mathematics member *Apr 2025 - Apr 2026*
- **IEEE**: IEEE Student & Geoscience and Remote Sensing Society (GRSS) member *Jan 2024 - Dec 2025*

JOURNAL/CONFERENCE REVIEWING

ALL REVIEWS ARE VERIFIED AND AVAILABLE ON THE **WOS/ORCID** PROFILE

- IEEE Transactions on Geoscience and Remote Sensing *ISSN: 1558-0644*
- Scientific Data *ISSN: 2052-4463*
- Sensing and Imaging *ISSN: 1557-2072*
- Biomedical Signal Processing and Control *ISSN: 1746-8108*
- Journal of Systems and Software *ISSN: 1873-1228*
- Scientific Reports *ISSN: 2045-2322*
- Discover Applied Sciences, formerly "SN Applied Sciences" *ISSN: 3004-9261*
- Journal of Computational Mathematics and Data Science *ISSN: 2772-4158*
- WSEAS Transactions on Business and Economics *ISSN: 1109-9526*
- Forum Geografi *ISSN: 0852-0682*
- Environmental Monitoring and Assessment *ISSN: 1573-2959*
- CMC-Computers, Materials & Continua *ISSN: 1546-2226*
- MDPI Informatics *ISSN: 2227-9709*
- IEEE International Conference on ICT Solutions for eHealth *ICTS4eHealth 2026*
- IEEE Symposium on Computers and Communications *ISCC 2026*
- ACM/SIGAPP Symposium On Applied Computing *SAC 2026*
- IEEE International Joint Conference on Neural Networks *IJCNN 2025*
- International Conference on Image Analysis and Processing *ICIAP 2025*
- International Academic Conference on Optics and Photonics *IACOP 2024*
- International Conference on Knowledge-Based and Intelligent Information & Engineering System *KES 2024, 2025*

May 27, 2026

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